

**MAKERERE UNIVERSITY
MAKERERE UNIVERSITY BUSINESS SCHOOL
FINAL EXAMINATION FOR THE BACHELOR OF BUSINESS COMPUTING
OF MAKERERE UNIVERSITY, ACADEMIC YEAR 2022/2023**

COURSE NAME: DATABASE DESIGN AND PROGRAMMING

YEAR OF STUDY: I

SEMESTER: II

TIME: 02PM - 04PM

COURSE CODE: BUC1224

DATE: Thursday 8th of June 2023

INSTRUCTIONS:

The paper is composed of section A and B

Answer all questions in both sections

ALL ANSWERS MUST BE WRITTEN IN YOUR ANSWER BOOKLET [ONLY SQL STATEMENTS].

The computer is to be used to help you in figuring out your answers.

SECTION A: (60 Marks) - Answer all questions using the HR account

Part I – 3marks each = 18marks

1. List all tables in the HR schema.
2. Display all columns of the employees table.
3. Display the last name, first name and salary of employees in the employees table.
4. Show the structure of the departments table.
5. List names of employees who are not working in department number 20
6. List street addresses and their postal codes in the locations table while giving each column a new heading of your choice.

Part II - 4marks each = 32marks

7. List employee names and their commission percentage. Only those employees who don't earn a commission should be returned.
8. Display employees whose first name ends with letter 'a' in descending order according to the first column.
9. Display the highest, average and lowest salary earned by employees.
10. Display all department names excluding those whose ids are 20, 30 and 40.
11. Display all department names whose number of characters is above 10.
12. List the name (first and last name) of employees whose salary earn between 2000 and 5000 sorted in ascending order according to first name.
13. Return employees, their salaries and commission. Return only those who don't earn a commission.

14. Return employees in department 10, 30, and 40 whose last name starts with 'A'.

Part III - 5 marks each = 10marks

15. List the first name, department name, and city of each employee. Sort the result in ascending order of the department name.

16. Display the first name, department id and name for all employees whether they belong to any department or not.

SECTION B (40 Marks) - Answer all questions

You must follow the instructions before answering this section.

Instructions

- a. Disconnect from the HR account if you're already connected. To do this, type **DISCONNECT** and press Enter on your keyboard
- b. Connect to the SYSTEM administrator account. To do this, run the code below:

CONNECT SYSTEM/hr

- c. Create an account for yourself. Use your last name as your username and your first name as the password. Use the code below as a guide to create your account:

**CREATE USER type_your_last_name_here IDENTIFIED BY type_your_first_name_here
DEFAULT TABLESPACE users TEMPORARY TABLESPACE temp QUOTA 500M ON users
PROFILE default ACCOUNT UNLOCK;**

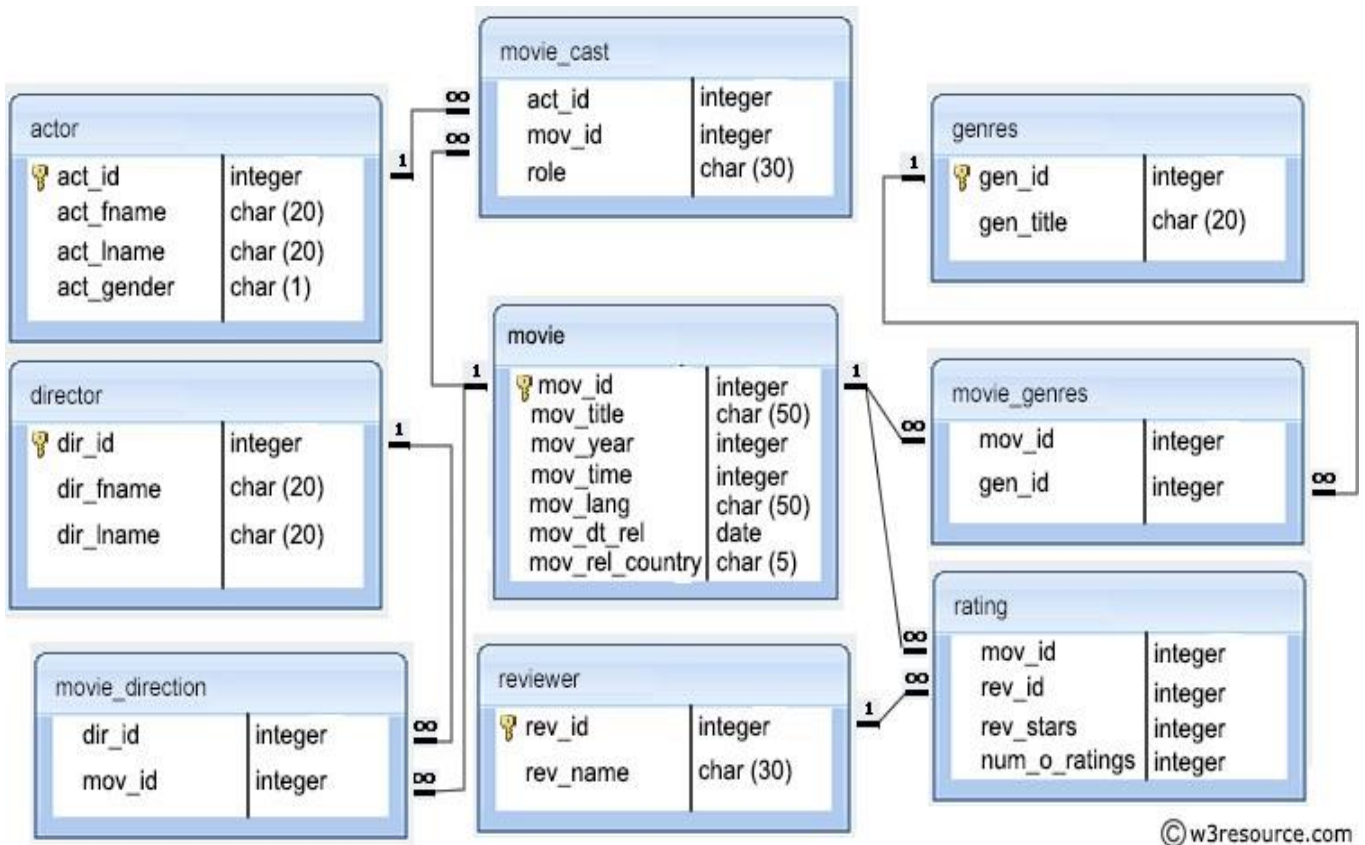
After successfully creating your account, run the code below:

GRANT CONNECT, RESOURCE TO type_your_last_name_here;

- d. Now disconnect from the **SYSTEM** account and connect to your new account. Use your account and the table below to attempt this section.

Additional Notes

Use the NUMBER data type for all columns with INT data type and VARCHAR2 data type for all columns with the VARCHAR and TEXT in the ERD below. You can as well use DATE data type instead of TIMESTAMP. The values in the brackets remain the same.



17. Create any 4 tables in the Movie Tracking System below. Enforce all necessary constraints (primary keys, foreign keys, checked columns, not null columns and unique keys) in the same CREATE statement. (16 Marks)
18. Add one column to any one table you created in 17. above. (2marks)
19. Delete one column from any one table you created in 17. above. (2marks)
20. Rename one column in any table you created in 17. above. (2marks)
21. Rename any two tables to their plural form for example from actor to actors. (4 Marks)
22. Insert at least three appropriate records into one of the tables you created (6 Marks)
23. Update any two records you inserted in question 22. above (4 marks)
24. Delete the third record you entered in 22. above from the table (2 marks)
25. Drop any one table from the database (2 marks)

End of Question Paper -